Alejandro Sánchez

AI Engineer & Data Scientist | Bioinformatics & Life Sciences

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SUMMARY

AI engineer and data scientist with a background in bioinformatics and biostatistics. 4+ years of research on sequencing and proteomics data, 3 peer-reviewed papers and a Master in Data Science & AI. I build machine learning models and LLM applications, evaluate them on real data and deploy them to production.

EXPERIENCE

Writer, Bioinformatics Expert (contractor) | Mercor, Remote **10/2026 – Present**
- Write expert-level bioinformatics and data analysis problems with worked solutions to train and evaluate AI models.
- Check that every solution is scientifically correct, verifiable and reproducible.

Bioinformatics Researcher | CBM Severo Ochoa (CSIC–UAM), Madrid **09/2022 – 07/2025**
- Led the computational side of multi-omics projects in J.M. Requena's group: Python, R and Bash pipelines for terabytes of Ribo-Seq, RNA-seq, WGS and proteomics data, replacing manual processing.
- Differential expression (DESeq2), genome assembly and proteogenomics on local servers and HPC (Slurm, GNU Parallel).
- Statistical analysis behind 2 peer-reviewed papers on genome annotation and proteogenomics (1 as first author).
- Main data contact for PIs and lab teams; supervised 5 students and technicians; managed sequencing providers.

Online Data Analyst (part-time) | Telus International AI, Remote **03/2025 – 06/2025**
- Evaluated search and AI model outputs against strict guidelines; produced ground-truth labels and documented edge cases.

Biotech Researcher | Universität Konstanz, Germany **04/2020 – 01/2022**
- Ran an environmental transcriptomics project end to end, from experimental design and bioreactors to data analysis.
- De novo transcriptome assembly on HPC (Slurm) and multivariate statistics in R; supervised 3 BSc/MSc students.

PROJECTS

Job Matcher – AI job application assistant | Personal project
- Multi-user app that analyses job offers in 5+ languages and writes tailored CVs and cover letters: Gemini extracts requirements with structured output, while matching and recommendations are deterministic, testable code, with validators against hallucinations.
- Chose the model with a hand-labelled evaluation on 20 real offers: **82/140 to 97/140** requirements correct. FastAPI, PostgreSQL, Docker and CI/CD on Oracle Cloud.

Sextant IA – LLM travel planner | Personal project, sextantia.eu
- Multi-stage Gemini pipeline with parallel calls, JSON repair, retries and a human-in-the-loop step; Django app live in production with CI/CD (245 commits).

Retail Demand Forecasting | Master's capstone, Nuclio Digital School **2025**
- Forecast weekly demand for 30K item-store series (6.7M rows) with Lasso and XGBoost, beating a naive baseline by **10% (RMSE)** and **13% (WMAPE)**.

EDUCATION

Master in Data Science & AI | Nuclio Digital School **2025**
MSc in Bioinformatics and Biostatistics | UOC–UB **2019 – 2022**
MSc in Microbiology | Universidad Autónoma de Madrid **2018**
Thesis at CBM Severo Ochoa (M. de Vega's group), published in Biomolecules (2020) as first author.
BSc in Biology | Universidad de Salamanca **2017**

SKILLS

AI & LLMs	Google Gemini API, LLM pipelines, prompt engineering, structured output, LLM evaluation, RAG (Google Search grounding)
ML & Statistics	scikit-learn, XGBoost, forecasting, clustering, hypothesis testing, Bayesian statistics
Backend & Cloud	Python, FastAPI, Django, SQL, PostgreSQL, Docker, GitHub Actions (CI/CD), Linux, Oracle Cloud
Data & BI	R, Bash, pandas, Power BI, Tableau, R Shiny
Bioinformatics	NGS (RNA-seq, Ribo-Seq, WGS), DESeq2, genome assembly, proteogenomics, HPC (Slurm)
Certifications	SQL (Advanced), HackerRank | ICH Good Clinical Practice E6 (R3)
Languages	Spanish (native), English (C1), German (B1), French (A2)
References	Dr. José M. Requena and Dr. Miguel de Vega (CBM Severo Ochoa); reference letters available